Variational Autoencoders

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Abstract

We investigate how latent space constraints impact generation quality, posterior collapse, and representation continuity in autoencoders. By comparing continuous (Standard VAE), modified-objective (InfoVAE), and discrete (VQ-VAE) architectures on CIFAR-10 and CelebA, we analyze reconstruction, latent utilization, and interpolation. Our findings reveal a fundamental trade-off between continuous manifold density and semantic sharpness, demonstrating that VQ-VAE's discrete structure provides the most robust balance for high-fidelity generation.

1 Introduction

Generative modeling aims to learn the underlying distribution of data such that new samples can be generated with similar characteristics. Variational Autoencoders (VAEs) (1) are a class of latent variable models that combine deep learning with probabilistic inference to learn structured representations of high-dimensional data. A comprehensive treatment of the theoretical foundations of VAEs can be found in (4; 5).

Definition 1.1 (Latent Variables). Latent variables z are unobserved random variables that explain the underlying structure of observed data x and are not directly available in the dataset.

Definition 1.2 (Latent Variable Models). A latent variable model defines a joint distribution $p(x, z)$ over observed variables and latent variables z, typically factorized as:

$$p(x, z) = p(z)p(x|z)$$

Definition 1.3 (Kullback-Leibler Divergence). The Kullback-Leibler (KL) divergence between two distributions P and Q is defined as:

$$D_{KL}(P\|Q) = \int P(x) \log \frac{P(x)}{Q(x)}\, dx$$

It is a non-negative measure of divergence and equals zero if and only if $P = Q$.

In latent variable models, learning requires maximizing the marginal likelihood: $p(x) = \int p(x, z)\, dz$.

However, this integral is often intractable due to the complexity of the posterior distribution $p(z|x)$.

Variational Autoencoders (1) introduce a variational distribution $q_\phi(z|x)$ to approximate the intractable posterior and optimize a tractable objective known as the Evidence Lower Bound (ELBO).

Theorem 1.1 (ELBO Decomposition). The log-likelihood of the data can be decomposed as:

$$\log p(x) = \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p(x, z)}{q_\phi(z|x)} \right] + D_{KL}(q_\phi(z|x)\|p(z|x))$$

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

Since the KL divergence is non-negative, it follows that $\log p(x) \geq \mathcal{L}_{\text{ELBO}}(x)$, where:

$$\mathcal{L}_{\text{ELBO}}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x) \| p(z))$$

2 Methodology

2.1 Variational Autoencoder (VAE)

The conditional distribution $p_\theta(x|z)$, parameterized by a neural network decoder (6), defines how observations are generated from latent variables. To approximate the intractable posterior $p(z|x)$, a variational distribution $q_\phi(z|x)$ is introduced, parameterized by an encoder network that outputs the mean μ and standard deviation σ of a Gaussian distribution.

The model is trained by maximizing the Evidence Lower Bound (ELBO) (1), defined as:

$$\text{ELBO}(x) = \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_\theta(x,z)}{q_\phi(z|x)}\right] = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x) \| p(z))$$

The first term corresponds to the reconstruction objective, while the second term regularizes the latent space to be close to the prior distribution. This objective enables learning of meaningful latent representations and allows generation of new samples by decoding latent vectors drawn from the prior.

A key component of VAE training is the reparameterization trick (1), which expresses the latent variable as:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

This formulation makes the sampling operation differentiable, enabling end-to-end optimization using gradient-based methods.

2.2 Vector Quantized VAE (VQ-VAE)

Vector Quantized Variational Autoencoders (2) differ from standard VAEs in their treatment of the latent space. While conventional VAEs (1) employ a continuous latent variable $z \sim \mathcal{N}(\mu, \sigma^2)$ and rely on KL divergence for regularization, VQ-VAE (2) introduces a discrete latent representation defined over a finite embedding space:

$$\mathcal{E} = \{e_k\}_{k=1}^K, \quad e_k \in \mathbb{R}^d$$

This discretization replaces stochastic sampling with a deterministic nearest-neighbor mapping, allowing the model to learn a structured and finite set of latent codes. As a result, the latent space becomes more interpretable and avoids issues such as posterior collapse commonly observed in continuous latent variable models (7).

Given an input x, the encoder produces a continuous latent representation $z_e(x)$, which is quantized via nearest-neighbor lookup:

$$z_q(x) = \arg\min_{e_k \in \mathcal{E}} \|z_e(x) - e_k\|_2$$

To enable gradient-based optimization, the stop-gradient operator $\text{sg}[\cdot]$ is applied to control the flow of gradients during backpropagation.

The training objective is given by (2):

$$\mathcal{L}_{\text{VQ-VAE}} = \underbrace{\|x - \hat{x}\|_2^2}_{\text{Reconstruction Loss}} + \underbrace{\|\text{sg}[z_e(x)] - z_q(x)\|_2^2}_{\text{Codebook Loss}} + \underbrace{\beta \|z_e(x) - \text{sg}[z_q(x)]\|_2^2}_{\text{Commitment Loss}}$$

- **Reconstruction Loss:** Ensures that the decoder reconstructs the input x from the quantized latent representation $z_q(x)$.
- **Codebook Loss:** Updates the embedding vectors by minimizing the distance between the encoder output $z_e(x)$ and the selected codebook vector $z_q(x)$. The stop-gradient operator ensures that only the codebook is updated.

- **Commitment Loss:** Encourages the encoder outputs to remain close to the assigned embedding vectors. The encoder is penalized for deviating from the selected codebook entries, with β controlling the strength of this constraint.

In contrast to standard VAEs (1), the KL divergence term is not required, as the latent space is explicitly constrained through discrete embeddings rather than probabilistic regularization.

Since the quantization operation is non-differentiable, gradients are approximated using the straight-through estimator (2), allowing gradients from the decoder to be directly propagated to the encoder during training.

2.3 InfoVAE

The standard ELBO objective (1) may lead to inaccurate amortized inference and can cause latent variables to be ignored when the decoder is sufficiently expressive. To address these limitations, **(author?)** (3) introduced a modified objective that explicitly controls the balance between data reconstruction and latent variable utilization.

The modified objective is defined as (3):

$$\mathcal{L}_{\text{InfoVAE}} = \mathbb{E}_{p_D(x)}\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - (1-\alpha)\, D_{KL}(q_\phi(z|x)\|p(z)) - (\alpha+\lambda-1)\, D(q_\phi(z)\|p(z))$$

where $q_\phi(z)$ denotes the aggregated posterior: $q_\phi(z) = \int q_\phi(z|x)\, p_D(x)\, dx$

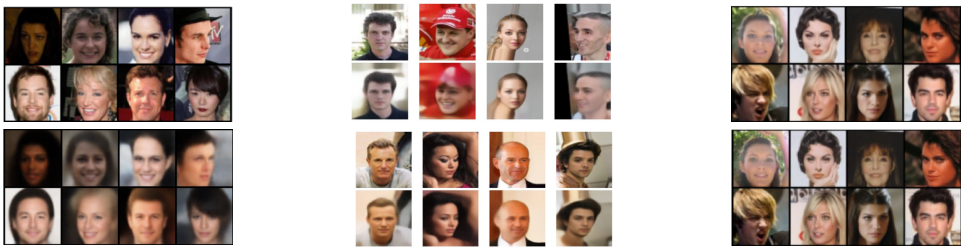
The parameter α controls the preference for mutual information between x and z, thereby encouraging the model to utilize the latent variables effectively. The parameter λ regulates the divergence between the aggregated posterior and the prior, influencing the structure of the latent space.

This formulation separates the objectives of accurate data reconstruction and correct inference, allowing improved control over the trade-off between them (3).

3 Experimental results

We conducted experiments to evaluate Standard VAEs (1), Vector Quantized VAEs (VQ-VAEs) (2), and Information Maximizing VAEs (InfoVAEs) (3) on CIFAR-10 and CelebA datasets. Section 3.1 compares reconstruction quality and generative performance, Section 3.2 analyzes posterior collapse using KL divergence and latent utilization, and Section 3.3 evaluates latent space interpolation and smoothness using LPIPS.

3.1 Latent Space Architecture Comparison

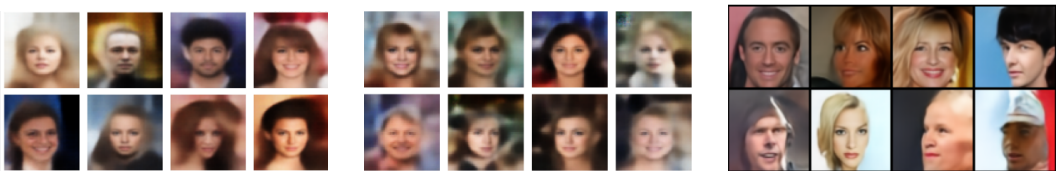


Standard VAE InfoVAE VQ-VAE

Figure 1: Reconstruction comparison on the CelebA dataset.(Top- Original, bottom - Reconstructed.

Model	Dataset	Reconstruction Loss	FID Score
Standard VAE	CelebA	0.0337	85.29
InfoVAE	CelebA	0.0756	68.21
VQ-VAE	CelebA	0.0028	18.18

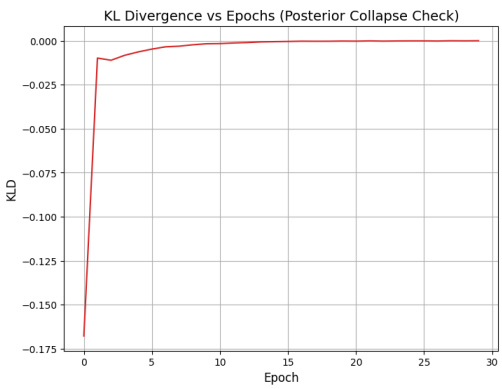
Table 1: Quantitative comparison of reconstruction accuracy and generative quality on the CelebA.



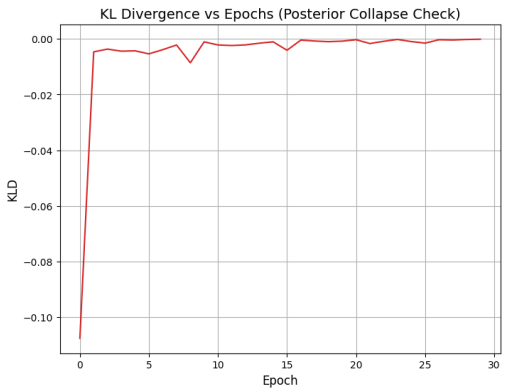
Standard VAE InfoVAE VQ-VAE

Figure 2: Qualitative comparison of generated samples on CelebA dataset.

3.2 Studying Posterior Collapse in VAEs



(a) 3-Layer ($\beta = 1.0$)



(b) 5-Layer ($\beta = 1.0$)

Figure 3: KL divergence trends across different decoder depths

Model Configuration	Active Units(Utilization)
3-Layer Decoder ($\beta = 1.0$)	0 / 32 (0.00%)
5-Layer Decoder ($\beta = 1.0$)	0 / 32 (0.00%)
3-Layer Decoder ($\beta = 0.00025$)	32 / 32 (100.00%)
5-Layer Decoder ($\beta = 0.00025$)	32 / 32 (100.00%)

Table 2: Summary of latent space utilization across different decoder depths and KL weights.

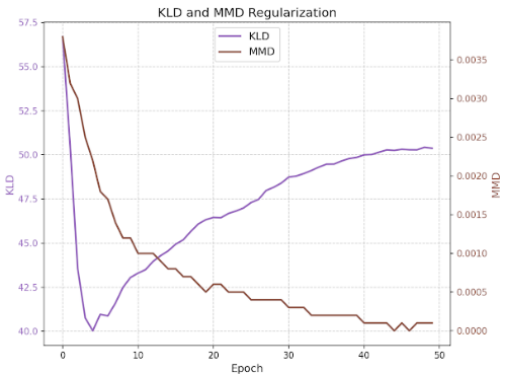


Figure 4: Training dynamics of InfoVAE showing KL-related term, MMD.

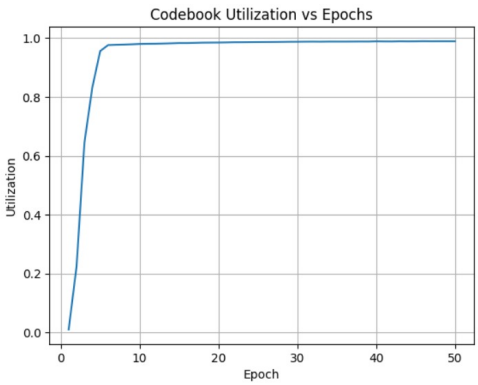
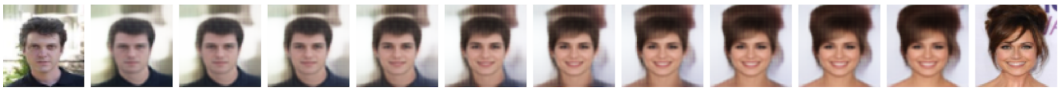


Figure 5: Codebook utilization of VQ-VAE during training.

3.3 Latent Space Interpolation and Representation Quality

(a) Interpolated images of Standard VAE



(b) Interpolated images of InfoVAE

(c) Interpolated imgaes of VQ-VAE

Model	Average LPIPS (Interpolation Smoothness)
Standard VAE	0.3106
InfoVAE	0.1832
VQ-VAE	0.0155

Table 3: Comparison of interpolation smoothness across models using average LPIPS. Lower values indicate smoother transitions.

4 Analysis and Conclusion

In this work, we evaluated the impact of continuous, discrete, and modified-objective latent constraints on Variational Autoencoders (1). Our comparison of Standard VAE (1), InfoVAE (3), and VQ-VAE (2) leads to the following observations:

- **Reconstruction and Fidelity:** VQ-VAE (2) produces the sharpest and most realistic outputs, achieving the best reconstruction loss and FID score. In contrast, Standard VAE (1) suffers from noticeable blurring due to Gaussian sampling, while InfoVAE (3) improves perceptual quality but still lacks the sharpness of VQ-VAE.
- **Posterior Collapse:** Standard VAE (1) shows severe posterior collapse under strong KL regularization, with very low latent utilization (7). InfoVAE (3) reduces this issue through MMD regularization, while VQ-VAE (2) avoids it entirely by using discrete latent representations.
- **Latent Space Interpolation:** Continuous models provide smooth transitions but often produce blurry results. VQ-VAE (2), although discrete, gives more stable and meaningful interpolations, supported by the lowest LPIPS score.
- **Overall Observation:** There is a clear trade-off between smoothness and sharpness. Continuous models (1) favor smooth latent spaces, whereas VQ-VAE (2) captures more meaningful and stable representations.

Overall, VQ-VAE (2) provides the best balance between reconstruction quality, latent utilization, and interpolation stability.

For more details, refer to our code on GitHub: GitHub Repository.

Acknowledgements

We would like to express our sincere gratitude to our mentor, **Varivashya Poladi**, for her valuable guidance, consistent support, and timely feedback throughout the project. Her insights greatly enhanced our understanding and progress on this project.

We also thank Professors **Aditya Gopalan** and **Shishir NY Kolathaya** for their valuable inputs and for creating a well-structured course that enabled this work.

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A Appendix

A.1 Derivation of the log-likelihood Decomposition

We begin with the marginal log-likelihood:

$$\log p(x) = \int q_\phi(z|x) \log p(x)\, dz = \mathbb{E}_{q_\phi(z|x)}[\log p(x)] \tag{1}$$

since $\log p(x)$ is constant with respect to z and $\int q_\phi(z|x)\, dz = 1$.
Now, using Bayes' theorem $p(x, z) = p(z|x)p(x)$, we write:

$$\begin{aligned}
\log p(x) &= \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p(x,z)}{p(z|x)} \right] && (2)\\
&= \mathbb{E}_{q_\phi(z|x)} \left[\log \left(\frac{p(x,z)}{q_\phi(z|x)} \cdot \frac{q_\phi(z|x)}{p(z|x)} \right) \right] && (3)\\
&= \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p(x,z)}{q_\phi(z|x)} \right] + \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{q_\phi(z|x)}{p(z|x)} \right] && (4)
\end{aligned}$$

We identify the two terms:

- **Evidence Lower Bound (ELBO):**

$$\mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p(x,z)}{q_\phi(z|x)} \right]$$

- **KL Divergence:**

$$\mathbf{KL}(q_\phi(z|x) \parallel p(z|x)) = \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{q_\phi(z|x)}{p(z|x)}\right]$$

Hence,

$$\log p(x) = \mathbf{ELBO} + \mathbf{KL}(q_\phi(z|x) \parallel p(z|x)) \tag{5}$$

Since KL divergence is always non-negative, it follows that:

$$\log p(x) \geq \mathbf{ELBO} \tag{6}$$

This completes the derivation.

$q_\phi(z \mid x)$: Encoder / Inference Model

The proxy for $p(z \mid x)$, which is the encoder distribution. It can be any directed graphical model, commonly parameterized using deep neural networks (6). For example:

$$(\mu, \sigma^2) = \text{EncoderNetwork}_\phi(x), \quad q_\phi(z \mid x) = \mathcal{N}(z \mid \mu, \text{diag}(\sigma^2))$$

This model is widely used due to its tractability and computational efficiency (1).

$p_\theta(x \mid z)$: Decoder / Generative Model

The proxy for $p(x \mid z)$, which is the decoder distribution or generative model. Like the encoder, it can be parameterized using deep neural networks (6). For example:

$$f_\theta(z) = \text{DecoderNetwork}_\theta(z), \quad p_\theta(x \mid z) = \mathcal{N}(x \mid f_\theta(z), \sigma_{\text{dec}}^2 I)$$

Here, σ_{dec} is a hyperparameter that may be fixed or learned (5).

A.2 Experimental Setup

In this section, we describe the experimental design, datasets, model configurations, and hyperparameters used in our study. We focus exclusively on the setup and methodology without discussing results or analysis.

A.2.1 Datasets

We conducted experiments on two benchmark datasets:

- **CelebA:** A large-scale dataset of aligned human face images, used to evaluate structured and high-resolution generative modeling.
- **CIFAR-10:** A dataset of low-resolution natural images with high variability, used to test model robustness under complex distributions.

A.2.2 Models Considered

We implemented and evaluated the following architectures:

- Standard Variational Autoencoder (VAE) (1)
- InfoVAE (modified objective with mutual information control) (3)
- Vector Quantized VAE (VQ-VAE) with discrete latent space (2)

All models were trained under comparable settings to ensure fair evaluation.

A.2.3 General Training Configuration

- Number of training epochs: 50
- Optimization: Gradient-based optimization using backpropagation (6)
- Reconstruction objective: Mean Squared Error (MSE)

A.2.4 Experiment 1: Latent Space Architecture Comparison

Objective: Compare the effect of different latent space formulations (continuous, discrete, and modified objective).

Setup:

- Models: VAE (1), InfoVAE (3), VQ-VAE (2)
- Datasets: CelebA
- Training duration: 50 epochs
- Consistent hyperparameters across models for fair comparison

A.2.5 Experiment 2: Posterior Collapse Analysis

Objective: Study the conditions under which posterior collapse occurs and how it can be mitigated (7).

Setup:

- Dataset: CIFAR-10
- Model: Standard VAE (1) with varying configurations
- Decoder depths:
  - 3-layer decoder
  - 5-layer decoder
- KL divergence weight (β):
  - $\beta = 1.0$ (standard ELBO)
  - $\beta = 0.00025$ (reduced regularization)
- Latent dimension: 32

Latent Utilization Criterion:

A latent dimension z_i is considered active if:

$$\mathrm{Var}_x(\mu_i(x)) > \epsilon$$

where $\epsilon = 0.01$.

A.2.6 Experiment 3: InfoVAE Configuration

- Dataset: CIFAR-10
- Latent dimension: 128
- Objective: Modified ELBO with additional divergence term (3)
- Regularization:
  - KL divergence term
  - Maximum Mean Discrepancy (MMD)
- Hyperparameters:
  - α: controls mutual information
  - λ: controls aggregated posterior alignment

A.2.7 Experiment 4: VQ-VAE Configuration

- Dataset: CIFAR-10
- Codebook size: $K = 512$
- Embedding dimension: $D = 64$
- Commitment loss weight: $\beta = 0.25$ (2)

A.2.8 Experiment 5: Latent Space Interpolation

Objective: Evaluate smoothness and semantic consistency of latent representations.

Setup:

- Dataset: CelebA
- Method: Linear interpolation between two latent vectors

Interpolation is performed as:

$$z_\alpha = (1-\alpha)z_1 + \alpha z_2$$

A.3 Detailed Evaluation Metrics

To maintain brevity in the main text, this section provides the comprehensive mathematical formulations for the evaluation criteria used across all experiments.

Reconstruction Loss (MSE): Measures pixel-level accuracy and is theoretically defined as the expected negative log-likelihood (1). In our practical implementation, this is calculated using Mean Squared Error across the pixel space:

$$\mathcal{L}_{rec} = \mathbb{E}_{q_\phi(z|x)}[-\log p_\theta(x|z)] \tag{7}$$

Fréchet Inception Distance (FID): Evaluates perceptual quality by comparing the distribution of generated samples to real images in the feature space of a pre-trained Inception-v3 network. A lower FID indicates a closer distributional match:

$$FID = \|\mu_r - \mu_g\|^2 + Tr(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}) \tag{8}$$

where (μ_r, Σ_r) and (μ_g, Σ_g) represent the mean and covariance of the real and generated image features.

Active Units (AU) and Latent Variance: A latent dimension is defined as an Active Unit if its KL divergence from the prior remains above a predefined threshold ($\epsilon = 0.01$) (7). This is quantified as:

$$AU = \sum_{i=1}^{L} I\left(\mathbb{E}_{x\sim p_{data}}[D_{KL}(q_\phi(z_i|x)\|p(z_i))] > \epsilon\right) \tag{9}$$

where L is the total number of latent dimensions and $I(\cdot)$ is the indicator function. Alternatively, for continuous latent models, an active unit z_k is one where the variance of its mean across the dataset exceeds $\epsilon = 0.01$:

$$Var_x(\mu_k(x)) > \epsilon \tag{10}$$

Perplexity (for VQ-VAE): Used to measure codebook utilization (2). Posterior collapse in discrete models manifests as under-utilization of the codebook:

$$Perplexity = \exp\left(-\sum_k p_k \log p_k\right) \tag{11}$$

Maximum Mean Discrepancy (MMD) (for InfoVAE): Tracks global regularization to ensure the aggregated posterior successfully aligns with the prior without enforcing individual posterior collapse (3):

$$MMD(q_\phi(z)\|p(z)) = \mathbb{E}_{q,q'}[k(z,z')] + \mathbb{E}_{p,p'}[k(z,z')] - 2\mathbb{E}_{q,p}[k(z,z')] \tag{12}$$

Learned Perceptual Image Patch Similarity (LPIPS): Quantifies interpolation smoothness. It uses deep features from a pre-trained network to measure perceptual differences, making it more aligned with human vision than pixel-wise metrics:

$$LPIPS(x,x') = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \|w_l \odot (\phi_l(x)_{hw} - \phi_l(x')_{hw})\|_2^2 \tag{13}$$

where x, x' are a pair of consecutive images along the interpolation path, ϕ_l are feature activations from the l-th layer of a pre-trained network, and w_l are learned weights.

A.4 Team members and Contributions

- **Abhinav:** Worked on VQ-VAE, implemented code and experiments, contributed to report writing and prepared the presentation.
- **Deviprasad:** Worked on Standard VAE and VQ-VAE; contributed to report writing and experiments.
- **Dhanunjaya:** Worked on InfoVAE and VQ-VAE; contributed to report writing and experiments.
- **Lokesh:** Studied InfoVAE and Standard VAE; contributed to report writing and experiments.
- **Swarup:** Prepared the presentation, worked on Standard VAE, and contributed to report writing.